Light

GCSE Computer Science — Data Representation: Numbers (Set 1)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Computers store and process data using only two digits.
What are these two digits?

- ○ **A.** 0 and 1
- ○ **B.** 1 and 2
- ○ **C.** 0 and 9
- ○ **D.** A and F

Q2 (1 mark)

In an 8-bit binary number, what is the place value of the leftmost bit?

- ○ **A.** 1

- B. 8
- C. 64
- D. 128

Q3 (1 mark)

What is the denary number 5 in binary?

- A. 0100
- B. 0101
- C. 0110
- D. 1010

Q4 (1 mark)

What is the binary number 1100 in denary?

- A. 6
- B. 10
- C. 14
- D. 12

Q5 (1 mark)

Hexadecimal is a base-16 number system.

Which letter represents the denary value 10 in hexadecimal?

- ○ **A.** F
- ○ **B.** B
- ○ **C.** E
- ○ **D.** A

Q6 (1 mark)

Programmers often use hexadecimal instead of binary.
Why is hexadecimal useful for representing binary data?

- ○ **A.** Hexadecimal numbers are smaller than binary numbers
- ○ **B.** Hexadecimal is easier for computers to process than binary
- ○ **C.** Hexadecimal uses fewer bits to store the same value
- ○ **D.** Hexadecimal is more compact and easier for humans to read than long binary strings

Q7 (1 mark)

How many different values can be represented using 4 bits?

- ○ **A.** 4
- ○ **B.** 8
- ○ **C.** 32
- ○ **D.** 16

Q8 (1 mark)

What is the result of adding the binary numbers 0011 and 0001?

- ○ **A.** 0010
- ○ **B.** 0100
- ○ **C.** 0101
- ○ **D.** 0011

Q9 (1 mark)

The binary number 00001010 is shifted one place to the left.
What is the result?

- ○ **A.** 00000101
- ○ **B.** 00010100
- ○ **C.** 00001011
- ○ **D.** 10100000

Q10 (1 mark)

The binary number 00001100 is shifted one place to the right.
What is the result?

- ○ **A.** 00011000
- ○ **B.** 00000110
- ○ **C.** 00001101

- ○ **D.** 00000011

Q11 (1 mark)

An unsigned 8-bit binary number can represent values from 0 to a maximum value.
What is this maximum value?

- ○ **A.** 127
- ○ **B.** 128
- ○ **C.** 255
- ○ **D.** 256

Q12 (1 mark)

A signed 8-bit binary number using two's complement can represent negative and positive values.
What is the range of values it can represent?

- ○ **A.** 0 to 255
- ○ **B.** -127 to 128
- ○ **C.** -128 to 127
- ○ **D.** -256 to 255

Q13 (2 marks)

Convert the denary number 73 to an 8-bit binary number.

- A. 01001001
- B. 01000111
- C. 01001011
- D. 01110011

Q14 (2 marks)

Convert the 8-bit binary number 10110101 to denary.

- A. 171
- B. 181
- C. 173
- D. 175

Q15 (2 marks)

Convert the binary number 11011110 to hexadecimal.

- A. DE
- B. ED
- C. DB
- D. BE

Q16 (2 marks)

Convert the hexadecimal number 4F to binary.

- ○ **A.** 01000111
- ○ **B.** 01001110
- ○ **C.** 01001111
- ○ **D.** 01011111

Q17 (2 marks)

Calculate the result of adding the binary numbers 01100101 and 00011010.

- ○ **A.** 01111110
- ○ **B.** 01111111
- ○ **C.** 10000000
- ○ **D.** 01110111

Q18 (2 marks)

The binary number 00000111 is shifted two places to the left.
What denary value does the result represent?

- ○ **A.** 14
- ○ **B.** 21
- ○ **C.** 28

- **D.** 35

Q19 (2 marks)

The binary number 01100000 is shifted three places to the right.
What denary value does the result represent?

- **A.** 6
- **B.** 12
- **C.** 24
- **D.** 48

Q20 (2 marks)

A student attempts to store the result of adding 200 + 100 in an unsigned 8-bit binary register.
What problem occurs?

- **A.** Underflow, because the result is too small
- **B.** Overflow, because the result exceeds the maximum value of 255
- **C.** The calculation works correctly with no issues
- **D.** The computer automatically uses 16 bits instead

Q21 (2 marks)

In 8-bit two's complement, what is the binary representation of -1?

- A. 00000001
- B. 10000001
- C. 11111110
- D. 11111111

Q22 (2 marks)

Convert the hexadecimal number 2B to denary.

- A. 27
- B. 43
- C. 45
- D. 53

Q23 (2 marks)

A computer system uses a 6-bit code to represent different characters. How many unique characters can be represented?

- A. 12
- B. 32
- C. 64
- D. 128

Q24 (3 marks)

Two unsigned 8-bit binary numbers are added together:
11001010 + 01110110
What is the correct denary result, and what problem occurs?

- ○ **A.** Result is 320, and overflow occurs because this exceeds 255
- ○ **B.** Result is 64, and no problem occurs
- ○ **C.** Result is 320, and no problem occurs because computers handle large numbers automatically
- ○ **D.** Result is 256, and underflow occurs

Q25 (3 marks)

Convert the denary number -45 to an 8-bit two's complement binary number.

- ○ **A.** 10101101
- ○ **B.** 11010011
- ○ **C.** 11010010
- ○ **D.** 10110011

Q26 (3 marks)

Convert the denary number 187 to hexadecimal.

- ○ **A.** BA
- ○ **B.** BB

- **C.** AB
- **D.** B7

Q27 (3 marks)

A signed 8-bit two's complement number 11110100 undergoes an arithmetic right shift of two places.
What is the result in denary?

- **A.** -3
- **B.** -12
- **C.** 61
- **D.** 253

Q28 (3 marks)

A student performs the following binary calculation:
01001110 + 00101101 + 00001011
What is the final result in denary?

- **A.** 130
- **B.** 132
- **C.** 134
- **D.** 136

Q29 (3 marks)

The 8-bit binary pattern 10010110 can be interpreted as either an unsigned integer or a signed two's complement integer.
What are the two possible denary values?

- ○ **A.** Unsigned: 150, Signed: -106
- ○ **B.** Unsigned: 150, Signed: -150
- ○ **C.** Unsigned: 106, Signed: -150
- ○ **D.** Unsigned: 106, Signed: -106

Q30 (3 marks)

A colour code is represented in hexadecimal as #A3F.
Assuming this is a 12-bit value (4 bits per hex digit), what is the equivalent value in denary?

- ○ **A.** 2623
- ○ **B.** 2591
- ○ **C.** 2687
- ○ **D.** 2559

Total: **55** marks • Time: **30 min**

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